

Question 1 - 0417/May/June/24/11 Q2

Complete each of the following sentences.

- (a) The scrambling of data being sent from one device to another device is called
..... [1]
- (b) Using a computer to maintain the temperature of a room is called
..... [1]
- (c) Unauthorised access to a computer system is called
..... [1]
- (d) A conversation between a systems analyst and a member of staff that is part of the analysis
is called
..... [1]

Question 2 - 0417/May/June/24/11 Q10

An autonomous vehicle uses an automated braking system. There is a sensor at the front of the vehicle to measure the distance between the vehicle and the obstacle. The brakes are applied when the vehicle gets too close to the obstacle.

Describe the use of a microprocessor in the automated braking system.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 3 - 0417/May/June/24/11 Q8

Virtual reality can be used when training for sports.

(a) State **two** other uses of virtual reality.

1

.....

2

.....

[2]

(b) Discuss the impact of virtual reality on everyday life.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Question 4 - 0417/May/June/24/12 Q10

Cars can have sensors at the front and rear of the vehicle to detect obstacles.

Describe how the sensor detects an obstacle and sends the data to the microprocessor when parking the car.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question 5 - 0417/May/June/24/12 Q4

Vehicles can contain microprocessor controlled devices to improve transport safety. For example when a vehicle approaches a road speed sign the microprocessor sends an alert to the driver.

Discuss the positive and negative effects of using microprocessor controlled devices in transport safety.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Question 6 - 0417/May/June/24/12 Q8

(a) Explain the differences between virtual reality (VR) and augmented reality (AR).

.....

.....

.....

.....

[2]

(b) Discuss the impact of augmented reality (AR) on everyday life.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Question 7 - 0417/Oct/Nov/23/12 Q11

A clothing retailer only sells its clothes on the internet.

(a) Discuss the benefits and drawbacks to the clothing retailer of using internet shopping.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[6]

(b) A customer is purchasing some new clothes from this retailer.

Describe the steps the customer would take when purchasing the new clothes using internet shopping.

[illegible]

[5]

Question 8 - 0417/May/June/23/12 Q13

Weather forecasters use computer modelling.

(a) Explain why computer modelling is used to forecast the weather.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Data is collected by different types of sensor and then input to the computer model to be processed.

Identify **two** sensors that could be used to collect weather data.

1

2

[2]

(c) The processed data is then output.

Identify **two** output devices that could be used.

1

2

[2]

Question 9 - 0417/May/June/23/12 Q14

ePublishing software is used to create an ePublication to display a school's yearly magazine.

(a) Describe the characteristics of an ePublication.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) The ePublishing software is subject to software copyright legislation.

Give **two** methods that could be used to prevent software copyright legislation being broken.

1

.....

2

.....

[2]

Question 10 - 0417/May/June/23/11 Q12

Many media companies now provide media streaming for users to watch their television programmes.

(a) Explain what is meant by the term media streaming.

.....

.....

.....

..... [2]

(b) Describe the benefits and drawbacks of media streaming.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 11 - 0417/May/June/23/11 Q8

- (a)** Compare Optical Mark Readers (OMRs) with Optical Character Readers (OCRs). Your answer must include similarities and differences.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[6]

- (b)** Identify **two** uses of Optical Mark Recognition (OMR).

1

2

[2]

Question 12 - 0417/May/June/23/13 Q3

A group of students are producing a class newsletter which they will email to parents of students in the class. They are deciding whether to use a laptop computer or a smartphone to type the newsletter.

- (a) Discuss the advantages and disadvantages of using a laptop computer rather than a smartphone for this purpose.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) The students have decided to use a smartphone. Data for the newsletter will be input, stored and output by the smartphone.
- (i) Identify **two** devices that are part of the smartphone that could be used to input the data for the newsletter.
- 1
- 2
- [2]

- (ii) Identify **two** devices that are part of the smartphone that could be used to output data from the newsletter.
- 1
- 2
- [2]

- (iii) Identify **two** appropriate devices that could be used to store data for the newsletter.
- 1
- 2
- [2]

- (c) The newsletter is to be saved as a PDF and sent as an email attachment to parents.
- Describe **two** benefits of saving the newsletter as a PDF.
- 1
-
- 2
-
- [2]

- (d) Describe the features that the students must include to make the newsletter appeal to the parents.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question 13 - 0417/Oct/Nov/23/12 Q6

Many teachers use multimedia projectors in their lessons.

- (a) State **two** health issues associated with the use of multimedia projectors.

1

2

[2]

- (b) State **two** safety issues associated with the use of multimedia projectors.

1

2

[2]

Question 14 - 0417/May/June/23/13 Q7

The retail industry uses point of sale (POS) systems.

(a) Discuss the benefits of using POS in the retail industry.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Describe the steps involved in automatic stock control when an item is sold at a POS terminal.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Question 15 - 0417/May/June/23/12 Q9

- (a) The products in a warehouse contain either a bar code or a Radio Frequency Identification (RFID) tag.

The data about each product is read electronically.

Compare bar code readers and RFID readers. Your answer must include similarities and differences.

[illegible]

[6]

- (b)** Identify **two** other uses of RFID.

1

2

[2]